It Takes Two to Tango: Using the Actor-Partner Interdependence Model for Studying the Coach-Athlete Relationship

Abstract

Research has repeatedly shown the importance of the relationship between a coach and an athlete. Despite these findings, we argue that the approaches used in previous research fail to sufficiently consider the dyadic and bidirectional nature of the coach-athlete relationship. Although the Actor-Partner Interdependence Model (APIM) was already proposed as a useful methodology for investigating and understanding the coach-athlete relationship, its potential has been underexploited in sport psychology. We discuss the possibly misleading conclusions that can be drawn from studies with an individual rather than a dyadic perspective and we propose to closely examine the dyadic pattern that can be inferred from the APIM, rather than making a priori assumptions about it. Practical implementations of the APIM in user-friendly apps are envisaged, as are relevant extensions of the APIM that allow to answer more advanced research questions by using more complex designs.

Keywords: Coach-athlete relationship, Actor-Partner Interdependence Model, Dyadic relationships

Serena Williams and her coach, Patrick Mouratoglou, have been working together since 2012, making their collaboration one of the longer-lasting ones in women's tennis. During these years, Williams not only won 10 grand slam titles, but she and her coach also built a strong relationship, as witnessed by the following quotes in Mouratoglou's book (2017):

> *"I always consider my players to be the best. I am convinced that they will become number one and I try to convey this energy to them."* (Mouratoglou, 2017, p. 168).

> *"When I think back to that first Wimbledon final that Serena and I won together, her expression remains forever engraved on my memory. I see again her joy and this immense happiness which made her so radiant. I remember her embrace when she climbed up into the stands to take me in her arms and share that magical moment with me."* (Mouratoglou, 2017, p. 202).

> *"The confidence that she gradually rediscovered in her game was reinforced by the presence by her side of a coach in whom she believed and who believed in her."* (Mouratoglou, 2017, p. 169).

Each of these quotes illustrate that the coach-athlete relationship may, at its best, satisfy, comfort, motivate or energize both coaches and athletes through very diverse interactions, making the coach-athlete relationship perhaps the most important dyadic relationship in sport (e.g., Côté & Gilbert, 2009; Lyle & Cushion, 2016). Jowett (2007) defined the coach-athlete relationship as "the situation in which coaches' and athletes' feelings, thoughts and behaviours are mutually and causally interconnected" and captured the dyadic nature of this relationship in the 3Cs framework in which commitment, closeness, and complementarity are essential (Jowett & Lavallee, 2007). Coaches and athletes do not stand on their own. They are inevitably interconnected, in need of each other, to achieve both individual and relational goals, to perform and excel, to reach their aspired successes and to actualize their potential (Horn, 1992; Jowett, 2017; Jowett & Shanmugam, 2016). The effectiveness of their relationship depends on characteristics of both the coach and the athlete, as well as on contextual factors (Horn, 1992).

While the quotes on the relationship between Williams and Mouratoglou refer to a professional coach-athlete relationship, similar phenomena may occur in semi-professional or recreational coach-athlete relationships that are often studied in sport psychology. The point we want to make here, however, is that the quotes of Mouratoglou clearly illustrate important dyadic features of the coach-athlete relationship. The first quote, for example, illustrates how characteristics of the coach (e.g., a coach's energy) may impact the performance or well-being of the athlete. The second quote illustrates how characteristics of an athlete (e.g., an athlete's emotions) may affect coaches. Taken together, the first and second quote serve as an illustration of the bidirectional nature of the coach-athlete relationship in which coaches' characteristics (i.e., their feelings, thoughts or behavior) affect athletes (the first quote) and athletes' characteristics (i.e., their feelings, thoughts or behavior) affect coaches (the second quote). The third quote further highlights that, athletes' *and* coaches' trust in each other can simultaneously influence the athletes' confidence. In other words, an athlete's outcome may simultaneously be affected by the athlete's own feelings, thoughts, behaviors and characteristics (i.e., *actor* effect) as well as by the characteristics, feelings, thoughts and behaviors of the coach (i.e., *partner effect*). Likewise, coaches' functioning can simultaneously be impacted by characteristics of coaches themselves (i.e., *actor effect*) and by characteristics of their athlete (i.e., *partner effect*).

In sum, to obtain a clear and comprehensive picture of the coach-athlete relationship, it may be important to acknowledge the bidirectional nature of the coach-athlete relationship and to consider both actor and partner effects. Considering the characteristics, thoughts, behaviors or feelings of both coaches and athletes, and considering both directions of the relationship will give us a more precise understanding of the dynamics that are at play in the coach-athlete relationship. Such a dyadic perspective on the coach-athlete relationship, however, requires deliberate choices and attunements on the methodological and statistical level.

This paper focuses on the application of the Actor-Partner Interdependence Model (APIM; Kenny, 1996; Kenny et al., 2006; Kenny & Cook, 1999) to investigate the coach-athlete relationship. The APIM is a dyadic model in which actor and partner effects can be estimated simultaneously on outcomes measured in both coaches and athletes. The advantages of using the APIM were already emphasized several times in sport psychology research (e.g., Gaudreau et al., 2020; Jowett, 2007, 2017; Jowett & Poczwardowski, 2007; Jowett & Shanmugam, 2016; Poczwardowski et al., 2002, 2006). Despite these pleas for the use of a more complete dyadic perspective, the approaches used in previous literature still largely ignore actor or partner effects and/or the bidirectional nature of the coach-athlete relationship. We performed a simple literature review in Web of Science, PubMed and SportDiscus in which the keywords '*coach-athlete relationship*' or '*athlete-coach relationship*' and '*actor-partner interdependence model*' or '*actor partner interdependence model*' or '*APIM'* were used. While this search may not retrieve all relevant publications, it gives a good indication. Using those terms, our search only yielded nine publications (i.e., Campo et al., 2022; Davis et al., 2013; Jackson et al., 2010, 2011; Jackson & Beauchamp, 2010; Jowett et al., 2013; Nicholls & Perry, 2016; Stebbings et al., 2016; Yang et al., 2015). There could be several reasons for the limited use of the APIM, such as the necessity to use a more complex design, the need for more sophisticated analyses and the additional methodological challenges (e.g., issues on confidentiality, data collection at different time point from individuals within a dyad) that need to be dealt with (Quinn et al., 2010; Sud, 2022). Furthermore, there might be limited awareness of the wealth of research questions that the APIM can address or a lack of understanding of the consequences of ignoring actor or partner effects and of taking a unidirectional rather than a bidirectional perspective.

The present contribution aims to address some of these issues and contains three parts. First, after introducing the APIM and explaining its implied dyadic perspective, we discuss two different studies that focus on the association between personality traits and relationship satisfaction. While those studies addressed similar research questions, they used a different methodological perspective. Building on those two studies, we point out the pitfalls of a priori excluding actor and/or partner effects and of using a single perspective of either athletes or coaches. We show how dyadic patterns can be tested in the APIM, and by using a hypothetical example we discuss the bias that can arise from dropping actor or partner effects. In the second part, we point to an easy-to-use app to perform APIM-analyses that may help overcome the technical barrier that some researchers may experience. This app is a web-based application that does not require any particular software and reports almost automatically the APIM analysis. In the final part, we provide a state-of-the-art overview of possible extensions of the APIM, which will enable researchers to broaden their horizons and to address research questions of increasing complexity.

The Actor-Partner Interdependence Model

As the name suggests, the Actor-Partner Interdependence Model is designed to evaluate interdependence within relationships. The model allows to consider both actor and partner effects simultaneously, for both dyad members. That is, for both coaches and athletes, the contribution of each of these effects on an outcome variable of interest can be estimated.

Before doing so, it is important to first evaluate the distinguishability of the dyad members. Distinguishability means that members of a pair can be discerned based on a characteristic or role they fulfill inside a dyad (Kenny et al., 2006). In the coach-athlete relationship the coach and the athlete take a distinct and unique role. Coaches are often expected to take the lead, make decisions, share knowledge and provide support and advice, while athletes are often expected to learn, perform, execute, communicate their own experiences and to receive the support and advice they need to face new challenges (e.g., Jowett, 2017). In contrast, in an indistinguishable dyad, both members have the same role and cannot be distinguished based on a specific characteristic. A same sex double pair in tennis would be an example of an indistinguishable dyad (e.g., Jackson et al., 2007). For a conceptual discussion on (in)distinguishability we refer the interested reader to Kenny et al. (2006) or Kenny and Ledermann (2010). Gistelinck et al. (2018) and Olsen and Kenny (2006) describe how to test indistinguishability and model indistinguishable dyadic data, respectively. In the remainder of this paper, we will focus on distinguishable dyads only.

Basic Components of the APIM

To investigate the bidirectional coach-athlete relationship in a cross-sectional design with the APIM, four variables of interest are required (see Figure 1). The two dependent or outcome variables, which are labelled $Y_{athlete}$ and Y_{coach}, stand for the outcomes of the athlete and coach (e.g., their well-being), respectively. The $X_{athlete}$ and X_{coach} predictor variables are measures of the athlete and the coach, respectively (e.g., their motivation). Those predictor variables may be correlated and are expected to explain a part of the variability in $Y_{athlete}$ and Y_{coach}.

The two most central components of the APIM are the actor effects and the partner effects. Actor effects (denoted by *a*) represent the effect of one's own predictor variable on one's own outcome variable and can thus be considered *intrapersonal* effects. In the APIM for distinguishable dyads, two potential actor effects are present: one for athletes (a_1) and one for coaches (a_2). In addition to these actor effects, the APIM includes partner effects (denoted by p). A partner effect is defined as the effect of one's own predictor variable on the outcome variable of one's partner. As such, the partner effects can be considered *interpersonal* effects because they represent an exchange between two individuals. In the example of the distinguishable coach-athlete relationship, two potential partner effects might be at play: on the one hand, there might be an influence from coaches on athletes (p_{21}) and on the other hand athletes might affect their coaches (p_{12}). As partner effects indicate that one's characteristics, thoughts, feelings or behaviors are dependent on characteristics of one's partner, they emphasize the relational ground on which relationships are built (Kenny, 2018; Kenny et al., 2006; Kenny & Cook, 1999). In addition, there is reciprocity in the relationship when both partner effects are present.

Written in the form of two straightforward linear regressions, the model can be summarized as:

$$Y_{athlete} = a_1 X_{athlete} + p_{21} X_{coach} + E_{athlete}$$

$$Y_{coach} = a_2 X_{coach} + p_{12} X_{athlete} + E_{coach}$$

Applied to our example, the first equation tells us that the well-being of the athlete, $Y_{athlete}$, is a function of the motivation of the athlete ($X_{athlete}$), the motivation of the coach (X_{coach}) and an error term $E_{athlete}$ (containing all the variability in $Y_{athlete}$ that is not explained by $X_{athlete}$ and X_{coach}). Here, a_1 represents the actor effect of the athlete, and shows the magnitude of the effect of the athlete's motivation on the athlete's well-being. The parameter p_{21}, on the other hand, captures the partner effect from the coach on the athlete and shows the magnitude of the effect of the motivation of the coach on the athlete's well-being. The second equation tells us that the well-being of the coach, Y_{coach}, is a function of the motivation of the coach (X_{coach}), the motivation of the athlete ($X_{athlete}$) and an error term E_{coach} (containing all the variability in

Y_{coach} that is not explained by $X_{athlete}$ and X_{coach}). Here, a_2 represents the actor effect of the coach, and shows the magnitude of the effect of the motivation of the coach on the well-being of the coach. The parameter p_{21}, on the other hand, captures the partner effect from the athlete on the coach and reveals the magnitude of the effect of the athlete's motivation on the well-being of the coach.

The APIM thus allows quantifying and comparing actor and partner effects for coaches and athletes and enables testing key issues within the coach-athlete relationship. Building on the above-mentioned hypothetical example in which the association between motivation and well-being is studied in coach-athlete dyads, several interesting research questions can be posed such as (a) can the athlete's well-being be explained by one's own motivation?; (b) can the athlete's well-being also be explained by the motivation of one's coach?; (c) which one of the first two mentioned effects has a greater contribution? In addition, it might be possible that researchers want to test a hypothesis positing a symmetrical relation, such as (d) is the effect of the motivation of the coach on the well-being of the athlete equal in size compared to the effect of the athlete's motivation in the prediction of the well-being of the coach? Similarly, if the relationship is conceived as unidirectional, it needs to be demonstrated that (e) one of the partner effects equals zero, while the other does not. In line with the asymmetrical power balance that often exists in coach-athlete dyads (Denison et al., 2017; Simpson et al., 2015; Stebbings et al., 2016), there might for instance be an effect of the coach on the athlete, while there is no comparable effect of the athlete on the coach. Clearly, the APIM allows for a more refined investigation of a wealth of phenomena within the coach-athlete relationship which might enrich our knowledge and understanding on the relationship.

To illustrate the added value of taking a dyadic perspective when using the APIM, we outline and discuss two contrasting publications (Jowett et al., 2012; Yang et al., 2015) that explored the association between personality traits and relationship quality. In the first study, Jowett and colleagues (2012) took an individual perspective to investigate the association between personality, relationship quality, perceptions of coach empathy and satisfaction with training in athletes. This implies that only athletes' personality traits and perceptions about the coach and the coach-athlete relationship were considered. Zooming in on the association between personality traits (agreeableness, emotional stability, extroversion, conscientiousness and intellect) and perceived relationship quality (closeness, commitment and complementarity), the authors found that athletes' agreeableness was positively associated with their own perceptions of relationship quality. However, as coaches and athletes are presumably influencing each other, the responses on a measurement (e.g., one's perception of the relationship quality) may not only reflect the characteristics of the person (e.g., one's own personality traits) but also the characteristics of the person's partner (e.g., one's partner personality traits) (Kenny & Cook, 1999). In other words, in the study of Jowett et al. (2012), the personality of the coach may also have influenced the athlete's perception of the coach-athlete relationship quality as well, over and above the influence of the athlete's own personality traits. Since the researchers had chosen to focus only on the characteristics and perceptions of the athletes, it is impossible – as also acknowledged by the authors – to explore this in more detail.

The second study, performed by Yang and colleagues (2015), investigated the same association between personality traits (neuroticism, extroversion, openness, agreeableness and conscientiousness) and relationship quality (closeness, commitment and complementarity). However, in this second study, a dyadic perspective was taken. That is, both the athlete and the coach were asked to report about their personality and their perceptions of the relationship quality. Interestingly, Yang et al. (2015) provided evidence that not only actor effects, but also partner effects were at play. Their research revealed actor effects for conscientiousness, extroversion and neuroticism on perceptions of the coach-athlete relationship for both coaches and athletes as well as a partner effect in which athletes' personality traits (i.e., conscientiousness, extraversion and neuroticism) were associated with the coaches' perceptions of the relationship quality. More specifically, in this study athletes' conscientiousness and extroversion scores were positively related to coaches' perceptions of relationship quality while athletes' neuroticism scores were negatively related to relationship quality. Such partner effects could never have been revealed if an individual perspective was adopted.

In comparison with the initial study of Jowett et al. (2012), the second study of Yang et al. (2015) illustrates how the dyadic investigation of the coach-athlete relationship leads to a more refined, holistic, and precise understanding of the dynamics within the coach-athlete relationship. We do not claim that such additional effects will be present in all settings and for all types of studied outcomes, but we argue that it may be misleading to assume that only one of the four possible effects (e.g., the actor effect for athletes out of the two actor and two partner effects) in this dyadic relation is at play. In the next paragraph, we discuss how different dyadic patterns can be explored when taking a dyadic perspective. Furthermore, we will show in the paragraph thereafter that falsely ignoring an actor effect or a partner effect may lead to erroneous conclusions about the effects of interest.

Different APIM Patterns in the Coach-Athlete Relationship

The APIM serves as an excellent starting-point to reveal *dyadic patterns* in the coach-athlete relationship. Dyadic patterns represent a specific configuration of actor relative to partner effects and tell us more about the importance of both types of effects. As the APIM includes both actor and partner effects for both coaches and athletes, dyadic patterns can easily be determined. Traditionally, four APIM patterns are discerned to differentiate between the configurations of actor relative to partner effects (Kenny, 2018; Kenny & Cook, 1999;

Kenny & Ledermann, 2010). Each of these four patterns imply different processes in the coach-athlete relationship (Kenny & Ledermann, 2010) and examining them will therefore deepen our understanding and lead to important insights about the dynamics between coaches and athletes.

The actor only pattern (or *actor-oriented pattern*) assumes an actor effect that is different from zero and a non-existing partner effect (i.e., equals to zero). Many studies in sport psychology presuppose an actor-oriented perspective, without assuming or testing partner effects. When Jowett et al. (2012), for example, demonstrated how athletes' personality traits were associated with athletes' perceptions of the coach-athlete relationship quality, the possible contribution of the personality traits of the coach on the athletes' perceptions of the relationship quality were, as already mentioned, ignored.

In a *partner only pattern* (or *partner-oriented pattern*) only partner effects are assumed, whereas the actor effects are expected to be non-existing (i.e., equivalent to zero). According to this pattern, coaches (athletes, respectively) are affected by athletes' (coaches', respectively) characteristics (i.e., their scores on X), but not by one's own characteristics. Lafrenière et al. (2011), for example, investigated how coaches' perceptions of their own autonomy supportive and controlling behaviors affected the athletes' perception of the coach-athlete relationship quality (i.e., the partner effect from coaches on athletes), while the athletes' perceptions of the coaches' behaviors were not considered (i.e., the actor effect of athletes). Here, the authors implicitly presupposed the actor effect to be non-existing.

A third pattern that can be discerned, is the *couple-oriented pattern* in which actor and partner effects are equal (or not significantly different). An athlete (coach, respectively) is, from this point of view, as much affected by one's own characteristics as by the characteristics of one's coach (athlete, respectively). Jackson and Beauchamp (2010), for example, investigated the effects of self-efficacy, other-efficacy and relation-inferred self-efficacy on relationship commitment, effort and satisfaction in coach-athlete dyads. They demonstrated how both one's own perception of other-efficacy and one's partner's perception of other-efficacy were positively related to one's own level of relationship commitment, and those effects were similar in magnitude.

Finally, in the *social comparison* or *contrast pattern,* actor and partner effects are assumed to be equal in absolute magnitude, but opposite in sign. Jowett et al. (2013), for example, investigated how harmonious and obsessive passion were associated with interpersonal conflict and relationship quality. They found that coaches' obsessive passion was negatively associated with coaches' relationship satisfaction, while athletes' obsessive passion was positively related to coaches' relationship satisfaction, with both effects similar in absolute magnitude.

The studies by Jowett et al. (2013) and Jackson and Beauchamp (2010) illustrate that, depending on the concepts of interest, different dyadic patterns can be found for interactions and phenomena in the coach-athlete relationship. As such, a clear understanding of the interplay between coaches and athletes can only be obtained when both coaches' and athletes' perspectives or characteristics are simultaneously considered. As another example, Davis et al. (2013) studied the association between attachment styles, the quality of the coach athlete relationship and relationship satisfaction from a dyadic perspective. Not only actor effects, but also partner effects of athletes' and coaches' anxious or avoidant attachment style on their perceived relationship quality or relationship satisfaction were found. Such findings are important to obtain a better understanding of the processes involved in the formation and maintenance of relational bonds between coaches and athletes. They provide researchers, athletes and coaches with more nuanced insights and practical implications on how to strengthen the bond between coaches and athletes. We want to emphasize here that it is not only important to consider actor and partner effects for outcomes that are dyadic in nature such as relationship quality, but also for more individual outcomes such as personal well-being. Stebbings et al. (2016), for example, examined potential processes of bidirectional well- and ill-being contagion between coach-athlete dyad members and found evidence that athletes' and coaches' pre-session well- and ill-being were both associated with changes in athletes' well- and ill-being over the course of the session, while the reciprocal transfer from athlete to coach was not fully supported. Although a limited number of studies now adopt the APIM in studying the coach-athlete relationship, the vast majority of available studies rely on a unidirectional perspective only. Yet, and as we will show next, a priori and erroneously assuming an actor only or a partner only pattern may lead to wrong conclusions about the true actor and partner effects, respectively, and hence lead to wrong conclusions about the dynamics of the coach-athlete relationship.

Consequences of Ignoring the Dyadic Ground of the Coach-Athlete Relationship

For the purpose of illustration, the consequences of a misspecified model in which researcher only consider actor or partner effects are illustrated by means of hypothetical examples. In the two fictive examples $X_{athlete}$, X_{coach}, $Y_{athlete}$ and Y_{coach} are assumed as standardized variables (i.e., with mean zero and variance equal to one). The predictor variable X and the outcome variable Y are thus supposed to be measured in both athletes and coaches. Furthermore, the correlation between $X_{athlete}$ and $X_{coach,}$ and the residual correlation between $Y_{athlete}$ and Y_{coach} (i.e., after accounting for $X_{athlete}$ and X_{coach}) are set to 0.5. Such mild-to-strong correlation between measurements is often observed in close dyads (Kenny et al., 2006).

In a first example (Figure 2, panel A), let us assume that the association between motivation and well-being is investigated. In this example, a couple-oriented pattern, in which actor and partner effects are equal, is assumed as the truth. The effect of one's own motivation (X variable) and one's partner motivation on well-being are both set to equal 0.30, which corresponds with a mild to strong effect, for both coaches and athletes (left panel A). However, if the researcher only considers actor effects (i.e., an actor pattern only) as shown in the upper right A-panel of Figure 2, then the estimated effect of one's own motivation on one's own well-being equals 0.45 instead of 0.30. That is, the true actor effects are over-estimated by 50%. Similarly, if the researcher would only consider partner effects (i.e., a partner-oriented only pattern) as in the lower right A-panel of Figure 2, the researcher would over-estimate the partner effects by 50% as well.

In a second example (Figure 2, panel B), a contrast pattern, in which actor and partner effects are equal in absolute magnitude but opposite in sign, is assumed as the underlying truth (left panel B). Let us assume that the coaches' obsessive passion (X_{coach}) is negatively associated with coaches' relationship satisfaction (Y_{coach}), while athletes' obsessive passion ($X_{athlete}$) is positively associated with coaches' relationship satisfaction (Y_{coach}) with standardized effects set to equal -0.30 and 0.30 (and the same effects for the athlete). If the researcher would only consider intrapersonal effects (i.e., an actor-oriented pattern) as shown in the upper right B-panel of Figure 2, then the estimated effect of one's own obsessive passion on one's own relationship satisfaction equals -0.15 instead of -0.30. That is, the true actor effect is under-estimated by 50%. Similarly, if the researcher would only consider partner effects (i.e., a partner only pattern) as in the lower right B-panel of Figure 2, the researcher would under-estimate the partner effect by 50% as well.

Here, only two examples are shown with many restrictions like equal effects amongst roles (i.e., equal effects for coaches and athletes), and equal or opposite actor and partner effects, but the conclusions below are more general, and also apply to situations with other values for the actor and partner effects. Clearly, researchers who focus on intrapersonal explanations when studying the athlete-coach relationship (i.e., only actor effects are considered whereas partner effects are ignored) may under- or over-estimate true actor effects.

It is only when partner effects are modelled in addition to actor effects that it is possibly to correctly capture relational phenomena. Similarly, researchers focusing on interpersonal effects who acknowledge partner effects but ignore potential actor effects, may under- or over-estimate the true partner effects. To grasp a more complete picture of relational phenomena in the coach-athlete relationship it is thus important to simultaneously consider actor and partner effects.

Is it also necessary to simultaneously model the outcome of the coach and athlete? Perhaps somewhat surprising and counter intuitive, the answer to this question is no. Separately modeling the outcome for coaches and athletes (but considering actor as well as partner effects in both models) does not introduce bias, nor does it lead to loss in efficiency (Kenny et al., 2006). Nevertheless, joint estimation allows for more possibilities such as more straightforward testing of distinguishability (e.g., testing whether actor and partner effects are similar for coaches and athletes), looking for dyadic patterns (e.g., testing if there is a couple-oriented pattern) and assessing the residual correlation in outcomes (i.e., testing how strongly the outcomes in athletes and coaches are correlated after accounting for the predictors).

In sum, to obtain correct estimates of the actor and partner effects we strongly recommend to take a bidirectional perspective, and to use the APIM. In the next section, we provide some practical guidelines for using the APIM.

Practical Guidelines for Using the APIM

Design

The use of a dyadic perspective has implications for the requirements of the study design and data collection. When applying the APIM as discussed above, a two-sided or reciprocal standard design is needed (Kenny et al., 2006). In such a configuration, there are *n* dyads and *2n* individuals. In the context of the coach-athlete relationship, data are thus collected in both coaches and athletes, which may make data collection more challenging.

Indeed, not only must both the coach and the athlete consent, they also have to respond both to a set of questionnaires involving the same concepts. This may result in less dyads with complete data (i.e., data of both the coach and the athlete) and the need for oversampling (Krasikova & LeBreton, 2012). Researchers will have to pay such price for getting a more complete picture of the coach-athlete relationship. Unfortunately, the extra financial costs, the extra time and resources that are needed in the dyadic approach may discourage researchers from taking such a perspective.

The natural next question is how many dyads are typically needed? Our earlier mentioned literature review on the use of the APIM for investigating the coach-athlete relationship revealed that the number of dyads in those studies ranged from 50 to 350 dyads, with an average of about 125 dyads. To determine more precisely the sample size needed to detect specific actor and partner effects in the APIM for a prespecified type 1 error and power level, an application, APIMPowerR[1] was created by Ackerman and Kenny (2016). Although a power study specifically tailored to the context of the research questions may be preferred, a sample of about 200 dyads can be used as a rough rule of thumb to detect medium actor and partner effects with sufficient power (Ackerman & Kenny, 2016).

Estimation of the Parameters of Interest in the APIM

Without going into much technical detail, it is important to briefly address the statistical techniques that can be used to fit the APIM. The APIM can be analyzed by either multilevel modeling (MLM) or structural equation modeling (SEM). MLM was specifically developed for the analysis of hierarchical data, such as a coach and an individual athlete (level 1) who are nested within a coach-athlete dyad (level 2). SEM on the other hand encompasses path analysis and allows to simultaneously fit several linear regression models, such as the two linear regression models discussed before. Although both MLM and SEM can be used for

[1] The app can be found on https://robert-a-ackerman.shinyapps.io/APIMPowerRdis/.

analyzing the APIM, each has advantages and disadvantages (for a detailed comparison see Kenny et al., 2006 and Ledermann & Kenny, 2017). When working with distinguishable dyads, the SEM-approach is often recommended (Kenny et al., 2006). Indeed, the path analysis approach is the most natural and straightforward formalization of the APIM and its different effects. As can be seen in Figure 1, the model can be viewed as a path analysis model and all the APIM-parameters can simultaneously be estimated with SEM-software. Furthermore, parameters can easily be constrained (for example, to test equality in actor effects between athletes and coaches) or functions of parameters (for example, the ratio of the partner effect to the actor effect) can easily be defined within the SEM-framework.

For researchers who are not familiar with SEM-software an easy-to-use app has been developed that does not require any software (Stas et al., 2018). *APIM_SEM*[2] is a free online program with a point-and-click interface that can be used for fitting standard or more complex APIMs. The app provides a graphical user interface and only requires a minimum of input information (i.e., a selection of the predictors and outcomes from the dataset). No prior knowledge of any (statistical) software is needed to use all features of the app. The program uses the package *lavaan* (Rosseel, 2012) in R to fit the SEM-models behind the scenes, but no installation of R is required. We will not provide a step-by-step guideline of the tool, but refer the interested reader to Stas et al. (2018) for a detailed description on how to use the app.

However, we highlight some of the nice features of *APIM_SEM*. In the app, the user has the possibility to graphically explore the data and to obtain summary tables and path diagrams with estimated actor and partner effects. The lavaan syntax that is used behind the scenes as well as full text sentences with meaningful interpretations can be obtained. Furthermore, *APIM_SEM* allows to explore the dyadic patterns in the data. To this end, the estimation of the parameter *k*, which is defined as the ratio of the partner effect to the actor

[2] The app can be accessed through the link http://lavaan.org/APIM_SEM/.

effect for each role (i.e., athlete and coach) is performed (Kenny & Ledermann, 2010). Although k can take any value in practice, researchers are particularly interested in a few specific values. If k equals 1, a *couple pattern* is detected, indicating equal partner and actor effects. In line with the hypothetical example, when the association between one's own motivation and one's well-being is the same as the association between one's partner's motivation and one's well-being, k equals 1. When k equals −1, a *contrast pattern* is observed, with actor and partner effects equal in size but opposite in sign. An *actor only pattern* occurs when k equals 0, indicating a partner effect of zero. For example, when the athletes' well-being only relates to their own motivation and not to their coaches' motivation, k equals 0. In general, for k to be defined, the actor effect must be nonzero to avoid division by zero. If the actor effect is equal to zero, a *partner only pattern* is observed. For every analysis performed with *APIM_SEM,* such patterns are tested.

Beyond the APIM

So far, we focused on investigating the relationship between an athlete and a coach in a cross-sectional design. That is, the antecedents (the X variables) and the consequences (the Y variables) in both the athlete and the coach were all expected to be measured at one timepoint. Such a cross-sectional design without experimentation can only show associations among constructs and therefore provides no information about potential causal links between them.

The APIM can also be applied in longitudinal settings in which two or more measurement points are used. If the same variables (e.g., motivation) are measured on two different occasions (e.g., pre-test and post-test, see Figure 3), the APIM will be equivalent to an auto-regressive cross-lagged model with two time points (Cook & Kenny, 2005). For example, a researcher can assess the effects of an athlete's motivation pre-test and a coach's motivation pre-test on an athlete's (or a coach's) motivation post-test. The actor and partner effects are then equivalent to stability and cross-lagged coefficients, respectively. If the actor effects for both athlete and coach are high, there is reliable stability in the degree to which they both feel motivated. If the two partner effects are positive, the athlete's pre-test motivation influences the coach's post-test motivation and the coach's pre-test motivation influences the athlete's post-test motivation. In other words, if both those effects are present, the influence process would be reciprocal. Depending on the dyadic pattern at play, this may imply different strategies to target the coach's or athlete's motivation.

Applying a pre-post design ensures that the predictive variable antecedes the outcome variable, but it does not yield definite evidence for attributing causality either. Cross-lagged panel models have further been criticized (Hamaker et al., 2015; Mund & Nestler, 2019) as they do not allow to differentiate between traits (which are stable over time) and states (which are occasion-specific). With three or more occasions, one should therefore use a cross-lagged panel model with random intercepts (see Hamaker et al., 2015 for more details) to distinguish for example between one's occasion-specific motivation and one's "usual" motivation.

When antecedents (variables X) and consequences (variables Y) are measured repeatedly over time in both dyad members in a longitudinal design, it becomes possible to address more challenging questions. Imagine a study in which athletes and coaches are assessed after five competitions during the season. After each competition, both the coach and the athlete report on their current motivation and well-being (Gaudreau et al., 2020). Several questions can be addressed in this setting, for example, (a) whether athletes experience higher well-being *when* their motivation at a particular timepoint is higher than usual (i.e., the *within*-actor effect in the athlete); (b) whether athletes *who* generally are highly motivated are more satisfied than athletes *who* generally are less motivated (i.e., the *between*-actor effect of the athlete); (c) whether coaches experience higher well-being *when* their athletes' motivation at a particular timepoint is higher than usual (i.e., the *within*-partner effect of the athlete on the coach); (d) whether coaches with athletes *who* generally are highly motivated are more satisfied than coaches with athletes *who* generally are less motivated (i.e., the *between*-partner effect of the athlete on the coach); and (e) whether coaches and athletes synchronously experience lower or higher well-being after each competition (moment-specific within-dyad correlation). While cross-sectional designs allow to address the *who* question (i.e., between-effects; e.g., it might tell us for example that athletes whose coaches are highly motivated, have a better well-being than athletes who have lowly motivated coaches), longitudinal designs can go a step further and answer the *when* questions (i.e., within-effects; e.g., it might tell us for example at which occasion the well-being of the athlete is higher). Again, such findings may have important implications on answering the question when and who to target in interventions. To the best of our knowledge, the APIM has never been applied this way in sport psychology to longitudinal data. Note that also for modelling a longitudinal APIM, an app, L-APIM[3] has been developed that much facilitates fitting such complex models (Gistelinck & Loeys, 2019). Alternatively, if interest lies in exploring the evolution of a construct of interest over time (e.g., how does the motivation evolve over the course of a competition for both the coach and the athlete, and how do these trajectories covary?), latent growth curve models for dyadic data may be used instead (Peugh et al., 2013).

Obviously the APIM may be extended in several other ways. One may consider multiple predictor variables that have actor and partner effects on the outcome of interest (e.g., both motivation and commitment may be predictive for one's well-being) or one may adjust the association of interest for one or more covariates. These covariates may be measured at the level of the individual (i.e., a within-dyad covariate; e.g., the gender of the coach or athlete), or at the level of the dyad (i.e., a between-dyad covariate; e.g., the duration of the athlete-coach relationship). When including additional covariates in the model,

[3] The app can be found at https://fgisteli.shinyapps.io/Shiny_LDD2/.

interesting new research questions may arise. For example, one might be interested in the comparison of the observed associations (e.g., in terms of actor and partner effects) between motivation and well-being in coach-athlete dyads with different gender combinations. In other words, one may compare APIMs in multiple groups (Ledermann et al., 2017). This is of particular interest, because, based on for example gender, different combinations of coach-athlete dyads exist. Indeed, when considering male (M), female (F) and non-binary (X) athletes and coaches, nine different combinations (i.e., M-M, M-F, M-X, F-M, F-F, F-X, X-M, X-F, X-X) are possible. Processes that are at play in the coach-athlete relationship may depend on such gender combinations (Jackson et al., 2011; Jowett & Nezlek, 2012). As a hypothetical example, if there would be strong partner effects of motivation on well-being in same-sex combinations, while no such partner effects can be found in other-sex combinations, these patterns may be left undetected when gender would be ignored in the standard APIM. Indeed, in the standard APIM one would estimate actor and partner effects that are averaged over all possible gender combinations, and thus important information would be missed. Similarly, some authors (Evans et al., 2017) have argued that future research on coach-athlete relationships should consider different sport types and performance level (e.g., competitive or not). The multi-group APIM will be an important tool to assess similarities and differences in actor and partner effects between those different groups.

Furthermore, one may want to investigate whether the actor and partner effects depend, for example, on the age of the athlete (i.e., Is the actor or partner effect moderated by age?). Not only such moderation analysis (Garcia et al., 2015) is possible within the APIM framework, but mediation questions can be addressed too (Ledermann et al., 2011). For example, Stebbings et al. (2016) explored whether the association between coaches' pre-training-session well- and ill-being and changes in athletes' well- and ill-being over the course of the training session were mediated by athletes' perceptions of their coaches' interpersonal style. Several easy-to-use tools to perform those more complex analyses are available within DyadR[4] (Kenny, 2019), a cluster of web applications that have recently been developed to support and help researchers in using and understanding those models.

A last extension of the APIM that is worth mentioning is the Group Actor-Partner Interdependence Model (GAPIM; Kenny & Garcia, 2012). So far, we focused on the relationship between a single athlete and a coach, but relationships between a team of sporters and their coach might be of interest as well. The GAPIM is a model that considers the interdependence among team members and allows to simultaneously study the effect of one's own characteristics on one's own outcome (the actor effect), the effect of the average characteristics of the other members on that outcome (the others effect), the effect of how similar one is compared to the other team members (the actor similarity effect), and how similar on average every other member is to its team members (the others similarity effect) (for a further discussion see Kenny & Garcia, 2012). Campo and colleagues (2022) demonstrated for example how not only one's own characteristics (e.g., one's own score on group identification) contributed to one's perception of the coach-created climate (the actor effect), but also how other team members' identification with the team affected one's perception of an empowering climate (the others effect).

As an alternative to the GAPIM, the one-with many model (OWM; Kenny et al., 2006) may be considered. The OWM model examines the associations between focal persons' and/or partners' features and a particular outcome. Within the OWM framework, focal persons are the persons of interest, whereas the partners are their dyadic partners. In the sports context the coach may, for example, be the focal person who is connected to several team members. In a reciprocal OWM design, focal persons provide reports about multiple partners and all the partners provide information about the focal person. For example, the coach rates

[4] All applications can be consulted at http://davidakenny.net/DyadR/DyadRweb.htm.

the athletes in the team, and each athlete rates one's coach on relationship quality. The OWM model may then be used to answer a variety of research questions, including (a) how much variability in the perception of relationship quality is due to differences between coaches or to difference between athletes within the coaches' team?; (b) whether coaches have a similar perception of their relationship quality with all of their athletes and vice versa; (c) whether there are unique components of the relationship not accounted by how the coach views all of their athletes and how all of the athletes view their coach. Brinberg et al. (2022) further extended the OWM model to the longitudinal setting.

Conclusion

This contribution focused on the methodological and statistical aspects of the investigation of the coach-athlete relationship. Coaches and athletes are interdependent individuals that pursue a common goal while facing challenges on their road to success. They share more than simply the time they spend together. Coaches and athletes also share investments, commitments, characteristics and interests, making them presumably more connected to each other (Kenny et al., 2006). We argued that this interdependency needs to be considered at both the level of the design and analysis.

This interdependency and the possible bidirectional nature of the coach-athlete relationship is at the very heart of the Actor-Partner Interdependence Model. Our strongest recommendation is that researchers do well to adjust the design and analysis of their study to take the inherent dyadic nature of the coach-athlete relationship into account. More specifically and despite the methodological and statistical challenges that need to be faced, data on the predictor and outcome variables of interest should be gathered among both members of the dyad whenever relevant.

Some of the published findings on the coach-athlete relationship that are based on an individual perspective may be elaborated on by taking a dyadic perspective. As shown in this paper, ignoring the bidirectional nature of the coach-athlete relationship, may result in under- or overestimation of both actor and partner effects. A re-evaluation in a dyadic framework of findings from earlier studies that took an individual perspective could therefore yield important new insights. We do not want to speculate about the conclusions of specific existing studies investigating the coach-athlete relationship, but we hope that this paper will convince more sports psychology researchers to replicate some of these studies by using a dyadic lens.

Importantly, the dyadic investigation of the coach-athlete relationship might also shed new light on practical interventions in the coach-athlete relationship. Depending on the study and the effects found for an association of interest, it might be more interesting that interventions target athletes or coaches or both. If only actor effects are found, interventions tailored at the level of the individual rather than the dyad may be enough. In contrast, when partner effects are present too, future interventions should consider the dyadic context.

To conclude, the APIM constitutes a promising approach that will help advance our understanding of the coach-athlete relationship in sport psychology research. The APIM may set the stage for future extensions and refinements that allow for more fine-grained insights in the complex dynamics between coaches and athletes. Multiple outcomes and predictors could be analyzed, longitudinal influences could be charted or dyadic relationships could be studied in team sports by using the GAPIM or OWM framework. Because easy-to-use tools are available for these different models, the barrier for researchers to overcome statistical challenges in the modelling and analysis of dyadic data is now fairly low too.

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[1] The app can be found on https://robert-a-ackerman.shinyapps.io/APIMPowerRdis/.

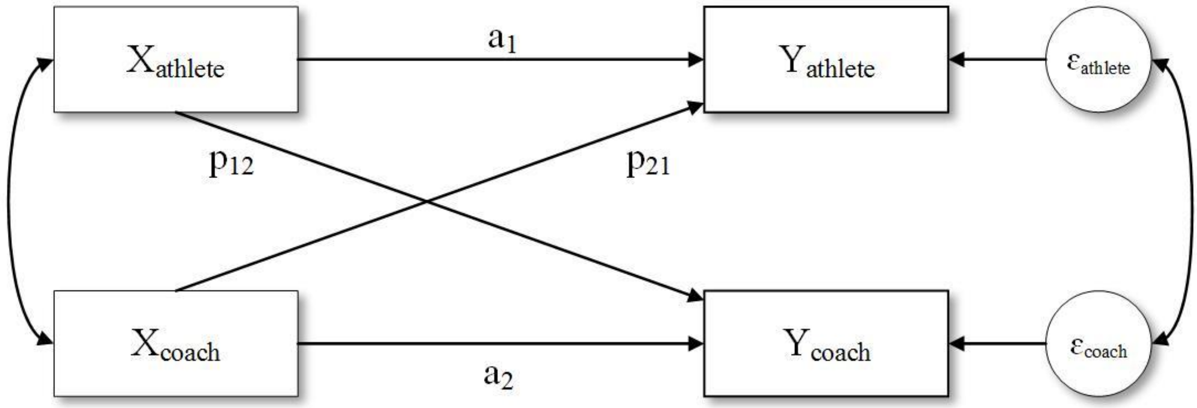
[2] The app can be accessed through the link http://lavaan.org/APIM_SEM/

[3] The app can be found at https://fgisteli.shinyapps.io/Shiny_LDD2/**.**

[4] All applications can be consulted at http://davidakenny.net/DyadR/DyadRweb.htm.

Figure 1

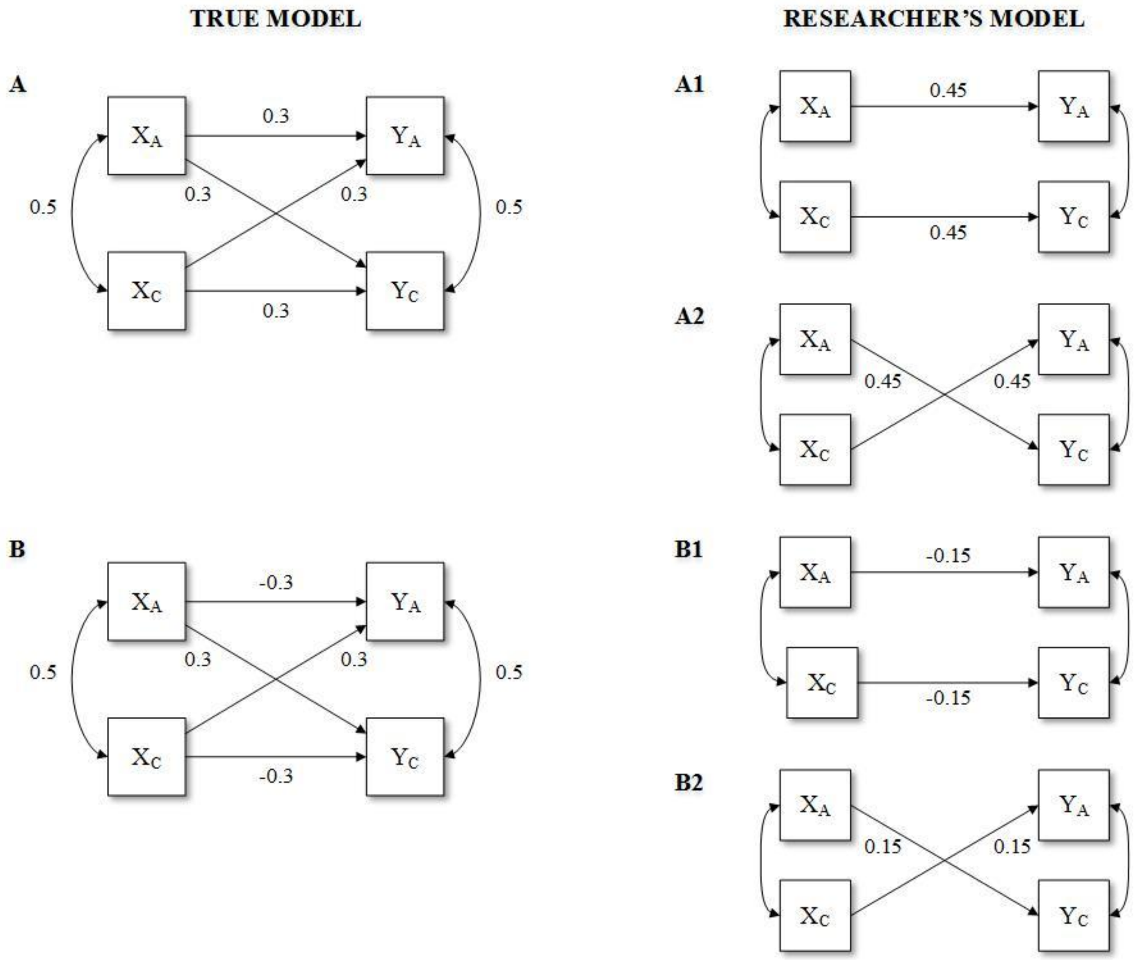
The Actor-Partner Interdependence Model for a Coach-Athlete Dyad



Note. The horizontal paths, denoted by *a*, represent the actor effects and the diagonal paths, denoted by *p*, represent the partner effects. The left double headed arrow represents the covariance between the predictor variables of coaches and athletes. The right double headed arrow represents a correlation between the error terms, reflecting the residual non-independence in the outcome scores that is not explained by the model.

Figure 2

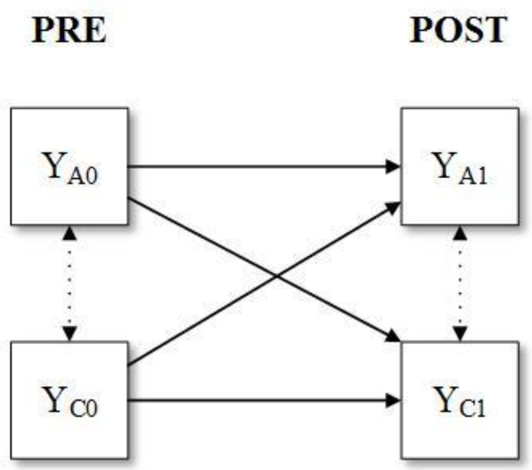
Biased Actor and Partner Effects due to Model Misspecification in Relationship Models



Note. In the left upper panel (panel A), a couple-oriented pattern is assumed to be the underlying truth. Both actor and partner effects equal 0.30. The right panel A1 represents the over-estimation of the actor effect when only considering intrapersonal effects. The right panel A2 represents the over-estimation of the partner effects when only considering the interpersonal effects into analysis. In the left lower panel (panel B) a contrast pattern is assumed to be the underlying truth. The actor effects equal -0.30, whereas the partner effects equal 0.30. The right panel B1 represents the under-estimation of the actor effects when only considering intrapersonal effects into analysis. The right panel B2 represents the under-estimation of the partner effects when only considering interpersonal effects into analysis.

Figure 3

The Application of the APIM in a Pre-Post Design



Note. The APIM applied to a pre-post design results in an auto-regressive cross-lagged model with two time point. Y_{A0} and Y_{A1} represent pre-test measures, whereas Y_{A1} and Y_{C1} represent post-test measures in athletes and coaches, respectively. The actor effects (horizontal paths) correspond with stability coefficients. The partner effects (diagonal paths) correspond with cross-lagged coefficients.